

EyeVision AI — Disease classifier evaluation

Model **dis-resnet50-odir-20261006-2119** (resnet50), trained 2026-10-06T22:40:05 on **odir**. Evaluated on: held-out TEST split (489 patients, 937 images; split by patient).

Research/educational AI estimate only. This system is not a medical diagnostic tool and does not provide a clinical diagnosis or eyeglass prescription. Please consult a qualified ophthalmologist/optometrist for diagnosis and refraction.

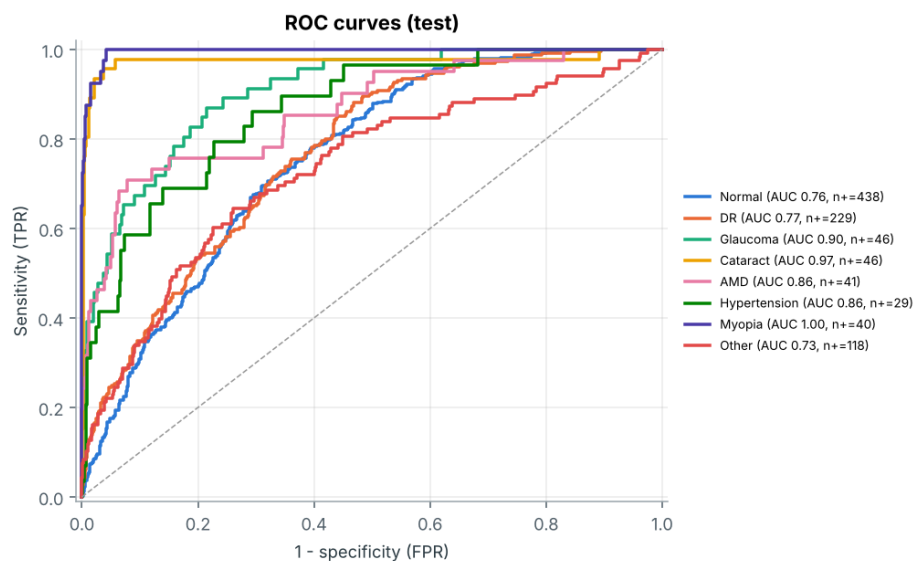
Summary (macro = mean over classes; micro = pooled over all labels)

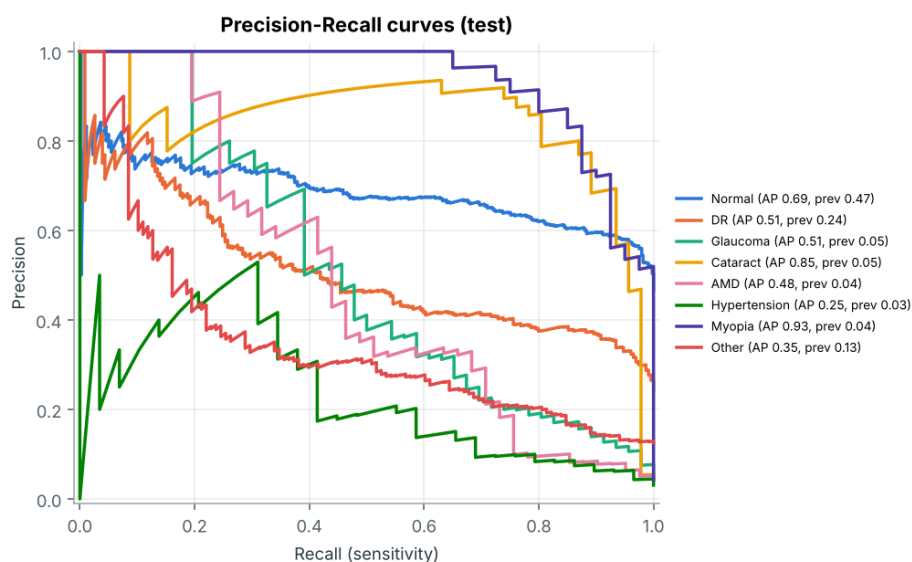
Metric	Macro	Micro
ROC-AUC	0.857	0.878
PR-AUC	0.571	0.511
F1	0.552	0.591
Precision	0.516	0.491
Sensitivity (recall)	0.644	0.742
Specificity	0.856	0.883

Per-class metrics (threshold tuned on validation set)

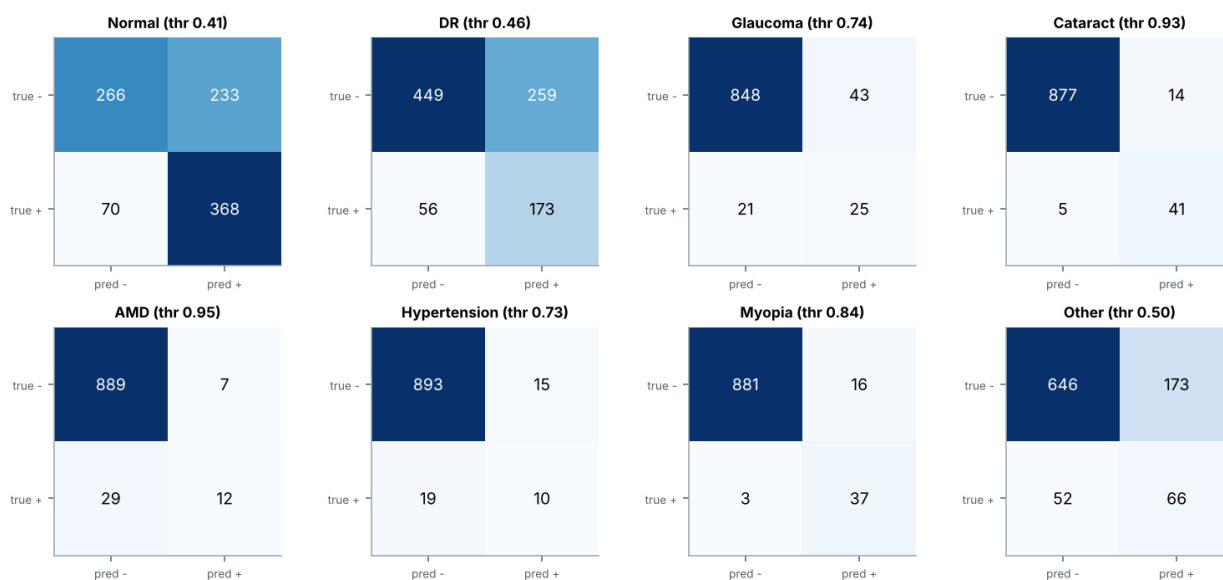
Class	n	n+	Thr	Prec	Sens	Spec	F1	ROC-AUC (95% CI)	PR-AUC
Normal	937	438	0.41	0.612	0.840	0.533	0.708	0.755 [0.72, 0.79]	0.685
DR	937	229	0.46	0.400	0.755	0.634	0.523	0.768 [0.74, 0.80]	0.511
Glaucoma	937	46	0.74	0.368	0.543	0.952	0.439	0.904 [0.87, 0.94]	0.510
Cataract	937	46	0.93	0.745	0.891	0.984	0.812	0.975 [0.93, 1.00]	0.848
AMD	937	41	0.95	0.632	0.293	0.992	0.400	0.863 [0.80, 0.93]	0.485
Hypertension *	937	29	0.73	0.400	0.345	0.983	0.370	0.863 [0.79, 0.92]	0.246
Myopia	937	40	0.84	0.698	0.925	0.982	0.796	0.996 [0.99, 1.00]	0.933
Other	937	118	0.50	0.276	0.559	0.789	0.370	0.730 [0.68, 0.78]	0.350

* fewer than 30 positive images — metric is unstable, interpret with caution. PR-AUC should be compared with the class prevalence (a random model's PR-AUC equals the prevalence).





Per-class confusion matrices (multi-label, one-vs-rest)



Comparison with simple baselines (same test patients)

Model	Macro ROC-AUC	Macro PR-AUC	Macro F1
resnet50 (this model)	0.857	0.571	0.552
Logistic regression (features)	0.712	0.320	0.363
Random forest (features)	0.716	0.322	0.346

Subgroup analysis (rows marked * are small; differences may be noise)

variable	group	n_images	n_patients	macro_roc_auc	macro_f1	macro_sensitivity	macro_specificity
Age group	40-59	422	219	0.847	0.476	0.617	0.834
Age group	60+	457	241	0.859	0.577	0.651	0.871
Age group	<40	58	29	0.904	0.644	0.833	0.854
Sex	F	398	210	0.871	0.571	0.664	0.868
Sex	M	539	279	0.845	0.523	0.633	0.847
Eye	L	471	471	0.855	0.526	0.636	0.859
Eye	R	466	466	0.859	0.575	0.650	0.853

variable	group	n_images	n_patients	macro_roc_auc	macro_f1	macro_sensitivity	macro_specificity
Source dataset	odir	937	489	0.857	0.552	0.644	0.856